TEXORA · LECTURE SHEET

Sewing Threads

Thread constructions from spun to core-spun and textured, the tex and ticket sizing systems, Z twist, and thread-consumption calculation with standard stitch factors — the arithmetic behind every thread order.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 3 · LEVEL 3 TERM 1

1. What a sewing thread must do

A sewing thread is an engineered yarn, not a leftover of weaving. It must survive being pulled through guides and a needle eye at up to 50–60 m/s of thread speed, form a clean stitch thousands of times, then spend the garment's life resisting abrasion, washing, light and load — all while adding as little bulk and cost as possible. The requirements are therefore: high **tenacity** with controlled elongation, low and even **friction**, **uniformity** (no slubs or thin places to snag in the eye), thermal stability against needle heat, and colour fastness matching the garment's care label.

2. Thread constructions

CONSTRUCTION	STRUCTURE	BEHAVIOUR AND TYPICAL USE
Spun (staple)	Short fibres twisted into yarn; cotton or polyester staple.	Soft, matt, good cover; weaker and hairier than filament. Cotton for natural-fibre garments and dyeable seams; spun polyester is the budget general thread.
Core-spun	Continuous filament core (usually polyester) wrapped with staple sheath (polyester or cotton).	Filament strength with a spun surface that protects against needle heat and sews kindly. The factory standard for shirts and trousers.
Continuous filament	Twisted multifilament nylon or polyester, often bonded.	Highest strength per size, smooth; for heavy duty, shoes, luggage; bonded types resist untwisting in multi-directional sewing.
Monofilament	Single translucent nylon filament.	Invisible hemming and quilting; stiff hand, can irritate skin — restricted uses.

Textured (bulk)	Air- or false-twist-textured filament with loops and bulk.	Soft, extensible, excellent cover in looper positions of overlock/coverstitch on knits; too weak and bulky as a needle thread for seams under load.

3. Thread size: ticket numbers and tex

Two systems run in parallel and must never be confused. **Tex** is direct: grams per 1,000 m — bigger number, heavier thread. **Ticket number** (metric ticket) is inverse: it derives from the cotton count system — bigger number, finer thread. The working conversions used in factories are:

Tex ≈ 1,000 ÷ metric ticket number · a tex 24 thread ≈ ticket 40; tex 40 ≈ ticket 25 (approx., by maker)

APPLICATION	TEX	TICKET (APPROX.)	NEEDLE NM
Fine shirting, blouses	18–24	55–40	65–80
Shirts, dresses (general)	24–30	40–30	75–90
Trousers, workwear	40	25	90–100
Denim, heavy seams	60–105	16–10	100–125
Loopers, overlock on knits	18–30 textured	—	—

Twist direction is almost universally **Z (final) twist** for machine sewing: the lockstitch hook action tends to open an S-twist thread, shedding strength and causing snarling, while Z twist is tightened by the same action. Twist level is a balance — too low frays; too high makes the thread lively, kinking into snarls at the needle.

4. Thread consumption: the calculation every technologist must do

Consumption is found from the **thread ratio (consumption factor)**: metres of thread used per metre of seam, for each thread in the stitch. The factors below are the standard planning values used throughout this course; they depend on stitch type, stitch density and material thickness, and the lockstitch value of **2.5** is the anchor figure.

STITCH TYPE	NEEDLE THREAD	BOBBIN / LOOPER THREAD	TOTAL PER METRE OF SEAM
301 lockstitch	1.25 m	1.25 m (bobbin)	**2.5 m**

401 chainstitch	2.0 m	3.5 m (looper)	5.5 m
504 overedge, 3-thread	4.0 m	10.0 m (two loopers)	14.0 m
516 safety stitch	—	—	19.5 m (401 + 504 combined)
607 flatlock, 4-thread	—	—	25.0 m approx.

Worked example — thread order for a trouser order. A basic trouser carries 6.2 m of 301 seaming, 3.4 m of 504 serging and 0.4 m of 401 chainstitch. Per garment: lockstitch thread = 6.2 × 2.5 = 15.5 m; overlock = 3.4 × 14.0 = 47.6 m; chain = 0.4 × 5.5 = 2.2 m; total = 65.3 m. Add 10% for waste, thread tails, repairs and machine threading: 65.3 × 1.10 = 71.8 m/garment. For an order of 10,000 trousers: 718,000 m ≈ **718 km** of thread. On 5,000 m cones that is 718,000 ÷ 5,000 = 143.6, so **144 cones** of the main thread (looper and needle threads are further split by their ratios where different threads are used — here the needle/looper split of the 504, 4.0:10.0, decides how many cones of textured looper thread to add).

> **WHY THE FACTOR IS NOT 2.0 FOR A LOCKSTITCH**
>
> A lockstitch seam of 1 m contains two threads, but each thread travels further than 1 m: it dives through the material at every stitch and the interlacing wraps around the opposing thread. At 4–5 stitches/cm on medium fabric the measured total settles near 2.5 m — the value factories plan with, verified by the pull-out test: sew a measured seam, unpick it carefully and measure the recovered threads.

5. Thread faults and storage

Thread-caused stoppages dominate sewing-floor losses: slubs and knots jam the needle eye; uneven twist snarls; low or variable lubrication raises friction and needle temperature; poor winding causes feed-off tangles. Buy to a specification (tex, construction, minimum breaking force, cone length, colour and fastness), test incoming lots by sewing, and store cones away from sunlight, dust and oil vapour — polyester is robust but its dye fades, and all threads pick up the sewing room's dust if left uncovered.

Glossary

TERM	MEANING
Tex	Direct thread size: mass in grams of 1,000 m of thread.

Ticket number	Inverse commercial size from the count system; larger ticket = finer thread.
Core-spun thread	Filament core wrapped in a staple sheath, combining strength with a heat-tolerant sewing surface.
Thread ratio	Metres of thread consumed per metre of seam for a given stitch type and density.
Z twist	Final twist direction standard for machine sewing threads, tightened rather than opened by hook action.

Key points and summary

- Core-spun polyester is the general factory standard; textured thread serves looper positions on knits; bonded filament serves heavy duty.
- Tex is direct, ticket is inverse — convert before comparing quotations (tex 24 ≈ ticket 40).
- Plan consumption with standard factors: 301 = 2.5 m/m, 401 = 5.5, 504 = 14.0, 516 = 19.5, plus about 10% waste.
- Thread quality is a productivity variable: slubs, snarls and poor winding stop machines more often than weak fabric does.

Practice

Answer in your own words; brief model answers follow.

1. A shirt uses 4.8 m of 301 seam and 1.6 m of 504 seam. Calculate the thread per garment, add 10% waste, and find the cones (5,000 m) needed for 25,000 shirts.
2. Convert: (a) ticket 30 to approximate tex; (b) tex 60 to approximate ticket. Which is the heavier thread, tex 24 or ticket 24?
3. Explain, from the hook's action, why machine threads are Z-twist.
4. A factory switches a trouser inseam from 401 to 301 to save thread. Calculate the saving per metre and state one sewing-quality risk of the change.
5. Why is textured thread specified for overlock loopers on jersey but not for the needle thread of a trouser seat seam?

Answer key

1. 301: 4.8 × 2.5 = 12.0 m; 504: 1.6 × 14.0 = 22.4 m; total 34.4 m; ×1.10 = 37.84 m/shirt. Order: 37.84 × 25,000 = 946,000 m ÷ 5,000 = 189.2, so 190 cones.
2. (a) Tex ≈ 1,000 ÷ 30 ≈ 33. (b) Ticket ≈ 1,000 ÷ 60 ≈ 16–17. Tex 24 is a defined weight (24 g/km); ticket 24 corresponds to about tex 40 — so ticket 24 is the heavier thread. The two numbers are not

comparable directly.

3. The rotating hook carries the needle loop around the bobbin in a direction that would untwist an S-twist thread, opening its plies and weakening it; the same motion tightens a Z-twist thread, keeping it compact through formation.
4. Saving = 5.5 − 2.5 = 3.0 m per metre of seam. Risk: the 301 seam extends only ~3–5% against the chainstitch's ~15–20%, so a stressed inseam may crack its thread; 301 also needs bobbin changes and cannot run back like a chain.
5. Textured thread is bulky, soft and extensible with excellent edge cover — ideal in loopers; but its breaking force is low and its bulk jams a needle eye under load, so a loaded seat seam needs a compact core-spun needle thread instead.

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